MILITARY SPECIFICATION

PUMP, VACUUM, COMPOUND, ROTARY, MOTOR DRIVEN, 12 CFM

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope: This specification covers one type of 12 CFM, electric motor driven, compound, rotary, vacuum pump for liquid oxygen tank servicing.

2. APPLICABLE DOCUMENTS

2,1 The following documents, of the issue in effect on the date of Invitation for Bids, form a part of this specification:

SPECIFICATION

FEDERAL

QQ-P-416	- Plating, cadmium (electro deposited)
TT-E-485	- Enamel, semi gloss, synthetic
TT-E-489	- Enamel, gloss, synthetic, for exterior and interior surfaces
PPP-B-585	- Boxes, wood, wirebound
PPP-B-591	- Boxes, fiberboard, wood-cleated
PPP-B-601	- Boxes, wood, cleated-plywood
PPP-B-621	- Boxes, wood, nailed and lock-corner
TT-P-636	- Primer coating, synthetic, wood and ferrous metal
CC-M-641	- Motor, alternating current (integral horse power)
TT-T-656	- Tricresyl Phosphate

MILITARY

JAN-P-100	- Packaging and packing for overseas shipment, general specification
MIL-P-116	- Preservation, methods of
MIL-C-490	Cleaning and preparation of ferrous and zinc coated surfaces for organic protective coatings
MIL-M-3171	- Magnesium, alloy, processes for corrosion protection of
MIL-E-4970	- Environmental testing, ground support equipment, general specification for

FSC 4310

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MIL-S-5002 - Surface treatments (except priming and painting) for metal and metal parts in aircraft
MIL-C-5541 - Chemical films for aluminum and aluminum alloys
MIL-C-6529 - Corrosion-preventive compound, aircraft engine
MIL-I-6866 - Inspection, penetrant method of
MIL-I-6868 - Inspection process, magnetic particle
MIL-A-8421 - Air transportability requirements, general specification for
MIL-S-8512 - Support equipment, aeronautical, special, general specification for the design of
MIL-P-8585 - Primer, zinc-chromate, for aircraft use
MIL-B-10377 - Box, wood cleated, veneer, paper overlaid
MIL-I-6181 - Interference limits, tests and design requirements, aircraft, electrical and electrical equipment

STANDARDS

FEDERAL

595 - Colors

MILITARY

MIL-STD-129 - Marking for shipment and storage
MIL-STD-130 - Identification Marking for U. S. Military Property

AIR FORCE-NAVY AERONAUTICAL

AN6235 - Filter Element, hydraulic replaceable, micronic line type
AN6236 - Filter Element, hydraulic replaceable, micronic reservoir type

PUBLICATION

AIR FORCE-NAVY AERONAUTICAL BULLETIN

143 - Specifications and Standards; use of

(When requesting specifications, standards and publications refer to both the title and number. Copies of' applicable specifications may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Penna., Attention, Code CDS).

2.2 Other Publications: The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of

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invitation for bids shall apply:

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

19.5:4 - Supplement on Instruments and Apparatus, Part 5, Chapter 4, Flow Measurement by Means of Standardized Nozzle-s and Orifice Plates

(Copies of the ASME power test code may be obtained from the American Society of Mechanical Engineers, 29 West 39th St., New York 18, N. Y.)

3. REQUIREMENTS

3.1 Preproduction: This specification makes provisions for preproduction testing.

3.2 Components: The vacuum pump shall consist of the following major components:

Description	See Requirement
(a) Power Plant	3.7
(b) Vacuum Pump	3.8
(c) Servicing Line	3.9
(d) Cabinet	3.10
(e) Base and Suspension	3.11
(f) Accessories	3.12

3.3 Selection of Specifications and Standards: Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with Bulletin 143.

3.4 Materials

3.4.1 Protective Treatment: When materials are used in the construction of the vacuum pump that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3,4,2 Gasketing and Insulating Materials: Rubber or similar gasketing and insulating materials shall be compounded to insure their resistance to the tricresyl phosphate pump lubricant and to withstand the environmental requirements.

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3.4.3 Combustible Material: Combustible materials shall be avoided in internal construction where high purity oxrity oxygen may concentrate during normal operation,

3.5 Design and Construction: The vacuum pump shall be designed and constructed in accordance with all applicable requirements of Military Specification MIL-S-85128 including the air transportability requirements, except where they conflict with the detail requirements of this document. In the event of a conflict the requirements of this specification shall govern.

3.5.1 The vacuum pump shall be an independent, self-contained unit designed for the purpose of evacuating the vacuum-tight annular insulating space of liquid oxyqen or nitrogen storage tanks to absolute pressures **Of** 10 microns Hg, or less.

3.5.2 Mounting: The vacuum pump shall be mounted on a sturdy base and enclosed in a cabinet suitable for providing adequate protection from environmental conditions. Insofar as practicable, all individual components shall be located and installed on the base to permit their being removed without removing other components.

3.5.3 Locking Devices: All screws, pins, bolts, et cetera, shall be equipped with locking devices. Safety wire, self-locking nuts, and cotter pins are acceptable.

3.6 Performance: The vacuum pump shall be capable of meeting the requirements specified herein.

3.6.1 Capacity

3.6.1.1 Free Air Pumping: The free air capacity of the vacuum pump shall be such that when it is operating at rated speed with both inlet and discharge connections open to atmospheric pressure, the quantity of air or other gases pumped through shall be in excess of 12 cfm measured at ambient temperature and pressure.

3.6.1.2 Reduced Pressure: The capacity of the vacuum pump at reduced pressures shall be such that when it is operating at rated speed in an ambient temperature of +70°F with the inlet pressure constant at 100 microns Hg, and the discharge open to atmospheric pressure, the quantity of air or other gases pumped through shall be not less than 11 cfm measured at inlet pressure and temperature conditions.

3.6.1.3 Ultimate Vacuum: When operating at rated speed in an ambient temperature of +70°F with the inlet completely blanked off except for a connection to an absolute pressure measuring device and the discharge open to atmospheric pressure, the vacuum pump shall be capable of attaining and holding an absolute pressure of less than 1.0 microns Hg.

3.6.2 Operating Time: The vacuum pump shall be capable of continuous

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operation under any of the conditions specified in paragraph 3.6.1 for periods of not less than 24 hours without requiring shutdown for servicing or maintenance.

3.6.3 Environmental Conditions

3.6.3.1 Temperature Operation: The vacuum pump shall be capable of operating in any ambient temperature between the limits of +15° and +130°F.

3.6.3.2 Temperature Exposure: The vacuum pump shall be capable of withstanding, without damage, storage at any temperature from -80° to +160°F.

3.6.3.3 Humidity: The vacuum pump shall not be damaged by operation or storage in any humidity conditions up to 95 percent relative humidity at 130°F

3.6.3.4 Fungus: The vacuum pump shall not be damaged by exposure to moist fungus encountered at locations in tropical and subtropical climates.

3.6.3.5 Salt Air: The vacuum pump shall not be damaged by exposure to or operation in an atmosphere containing salt-laden moisture.

3.6,3.6 Rain, Sleet, and Snow: The vacuum pump shall not be damaged by storage or operation under adverse conditions of rain, sleet, or snow.

3.7 Power Unit: The power unit shall consist of an electric motor and "Vee" belt drive system together with the controls, instruments and accessories needed for proper and safe operation.

3.7.1 Electric Motor: The motor shall be a 220 volt, 50-60 cycle, 3 phase drip proof unit conforming to Federal specification CC-N-641, The motor size shall be such that the driven load will not exceed the continuous duty rating of motor when the pump is operated under all design conditions and at rated capacity *on* 60 cycle current. The motor continuous duty rating when operating on 50 cycle current shall exceed the power consumed by the pump when operated 5/6th of rated speed.

3.7.2 Motor Controls: All electric motor controls and protective devices required for starting, stopping, overload protection, and under voltage protection shall be provided, A phase sequence control shall also be provided to prevent the pump from operating in the wrong direction, The phase sequence control shall either automatically reverse the phase sequence when the flexible cable is connected to an electrical power source in the wrong phase sequence or a separate manual phase reversing switch shall be provided on the pump assembly which will permit reversing the phase sequence without reconnecting the flexible cable.

3.7.3 Electric Power Cable: A 50-foot flexible 4-conductor, rough duty cable of adequate capacity shall be provided, the cable shall be permanently

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attached to the pump mounted electrical controls and grounded to the pump assembly. The unattached end of the cable shall have a suitable polarized male connector attached. The cable shall be stored on brackets on the exterior of the pump enclosure.

3.7.4 Radio Interference Suppression: The vacuum pump components shall be so constructed and shielded that radio interference generated during operation shall not exceed the limits established in specification MIL-I-6181,

3.8 Vacuum Pump: The vacuum pump shall be a two stage, compound, rotary type and shall be balanced to operate with minimum vibration. The pump shall be equipped with a gas ballast valve and air stripping to permit rapid pump oil drying and the ability to pump a high volume of condensibles. The vacuum pump shall be designed so that all critical parts are aligned with dowels and/or machined shoulders or pilots so that the pump can be assembled by semi-skilled personnel unfamiliar with vacuum pump maintenance. All -shaft seals shall be of inorganic materials suitable for extended life in the presence of the pump lubricant, The pump shall be designed so that it can be disassembled to the maximum practical amount for repair without disturbing critical component relationships such as the rotor and pump body. Shaft seals shall be replaceable from the pump exterior without further disassembly of the pump.

3.8.1 Pump Base: The vacuum pump and drive motor shall be mounted on a sturdy steel channel base. The base shall be of sufficient thickness and section to accurately position the pump and motor and shall prevent any stress or distortion to the pump body due to cabinet base distortion and/or vibration.

3.8.2 Lubricant: The vacuum pump shall be capable of meeting the performance requirements specified herein when lubricated with commercial grade vacuum pump oil having a total vapor pressure of less than 1 micron Hg at +100°F.

3.8.2.1 The vacuum pump shall be designed so that it will operate, although not necessarily in compliance with all absolute pressure requirements specified herein, for extended periods when lubricated with type 2 tricresyl phosphate conforming to TT-T-656. The normal lubricant in field service shall be type 2 tricresyl phosphate,

3.8.2.2 Lubricating System: The vacuum pump shall be provided with an automatic lubrication system which will insure an adequate lubricant supply to all parts requiring same under the operating conditions specified herein.

3.8.3 Cooling: The pump shall be air cooled and capable of maintaining proper operating temperature during operation in all environmental conditions specified in section 3.6.3. No auxiliary cooling system shall be permitted.

3.8.4 Separator: The vacuum pump shall incorporate features designed to effectively recover the lubrication fluid from the exhaust gases, separate condensible vapors from the lubricating fluid, and provide for discharging

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the condensable vapors from the pump. The lubricating fluid storage volume shall be provided with a drain shut-off valve and a drain tube or pipe that directs the draining pump lubricant away from any painted surface, Pump lubricant filler neck shall be at least 1-3/4 inches inside diameter. The pump lubricant storage volume shall have provisions for air stripping. These shall consist of provisions to admit up to 50 cubic feet per hour of dry nitrogen gas evenly across the low point of the lubricant storage volume. The connection shall consist of a 1/4 inch outside diameter metal tube for attachment of a 1/4 inch inside diameter rubber tube.

3.8.5 Inlet Connection: The vacuum inlet connection shall be a 1-1/2 inch internal pipe thread or larger, or a 1-3/4 inch inside diameter straight thread or larger.

3.9 Inlet Piping: The vacuum pump shall be equipped with an oil check device, a manual blank-off valve, a thermocouple vacuum. gage, an inlet filter, and flexible service lines.

3.9.1 Oil Check Device: An automatic device shall be provided on each pump assembly which will prevent flow of the vacuum pump lubricant through the servicing line and into the container being evacuated in case the pump stops.

3.9,2 Blank-Off Valve; A manual blank-off valve shall be provided. The blank-off valve shall be designed for 1-1/2 inch I.D., or larger, piping or tubing.

3.9.3 Thermocouple Vacuum Gage: A thermocouple vacuum gage shall be provided. The vacuum range shall be from one (1) micron to 5000 microns, The gage shall be calibrated against a standard to permit replacement of the thermocouple tubes without recalibrating the gage. The gage shall receive its energy from the vacuum pump electrical power. The thermocouple gage shall be installed so that it can read pump blank-off pressure or system pressure up-stream of the closed blank-off valve. The thermocouple gage shall be mounted on shock mounts to prevent damage due to the normal rough handling received by military equipment.

3.9.4 Inlet Filter: The vacuum pump shall be equipped with a replaceable-element-type filter between the servicing line and vacuum pump inlet to prevent passage of insulating material into the vacuum pump from the liquid-oxygen storage tank annular space being evacuated.

3.9.4.1 General Filter Design: The inlet filter shall be designed for solid and secure installation within the vacuum pump enclosing cabinet, so that it can be readily opened for inspection and replacement of elements, and so that it will seal vacuum-tight upon reassembly. A fitting suitable for vacuum-tight connection of the large diameter servicing line shall be provided in the servicing line filter element,

3.9.4.2 Filter Size: The inlet filter shall be of sufficient size and

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shall use a sufficient number of individual elements to insure free flow of gases into the vacuum pump from the container being evacuated.

3.9.4.3 Filter Elements: The inlet filter shall be designed to use replaceable elements conforming to AN6235 or AN6236.

3.9.5 Vacuum Hose: Two 72 ± 6 inch heavy duty flexible vacuum hoses suitable for use at absolute pressures of 10 microns Hg or less shall be supplied. One hose shall be at least 1-5/8 inches inside diameter and shall exceed the inside diameter of the pump end inlet connection, paragraph 3.8.5. The second hose shall be 1-inch inside diameter. The l-inch inside diameter hose shall have an adapter to permit coupling to the filter inlet fitting.

3.9.6 Couplings: The 1-inch I.D. hose shall be supplied complete with an end coupling in accordance with Figure la. The larger hose shall be supplied with an end coupling in accordance with Figure lb. The pump shall be supplied complete with the 1-inch I.D. hose installed. An instruction plate shall be attached to the pump in a prominent location indicating that the larger hose will provide faster pump down and can be used on 150-gallon and larger tanks procured or modified since 1958.

3.9.6.1 Coupling Cover: Each hose coupling shall be supplied with a polyethylene protective cover. The cover shall completely cover the sealing surface and the outside edge of the coupling. The cover shall snap in place and stay attached until removed. The cover shall be attached and removed by hand without tools. The cover shall be affixed to the coupling by a 12-inch length of 1/16 inch diameter, 6x7 stainless steel cable attached on each end by non-removable compression fittings.

3.10 Cabinet: A sturdy cabinet for containing and providing protection for the vacuum pump components, accessories, controls, and instruments shall be fabricated onto the base. The cabinet shall be constructed so that it, or applicable sections, can be readily removed from the vacuum pump to facilitate maintenance on or removal of components.

3.10.1 Cabinet Design: The cabinet shall be designed and constructed to protect the items installed within from adverse weather conditions, The top of the cabinet and the tops of all protuberances therefrom shall be shaped and slanted to resist the retention of rain, snow, or sleet. Panels designed to open for operating the vacuum pump shall be given particular attention in this respect so that snow or ice cannot lodge thereon in such a manner as to prevent ready access or actuation.

3.10.2 Size: The cabinet shall be of sufficient size to provide access to vacuum pump components for maintenance and repair. Space shall be provided for storing the servicing line within the cabinet when not in use.

3.10.3 Access Doors: The cabinet shall either have provisions for completely free access through three doors, each equal in size to the cabinet side or shall be hinged so that the roof and three sides can be swung up

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through 135° and secured. Doors, if used, shall be hinged horizontally along their top edge and shall rotate through 180°. The doors shall have provisions securely holding in the open as well as the closed position.

3.11 Mounting: The pump assembly including the pump and motor base shall be attached to a sturdy base. The base shall form the cabinet bottom and suspension support. The base shall be formed of 16 gauge steel, reinforced with suitable structural shapes under the pump base attachment points and around the outside edges.

3.11.1 Suspension: The chassis or base shall be suspended on two wheels and a landing strut. The wheels shall be heavy duty industrial type equipped with roller bearings and 12" x 4.00 heavy duty semi-pneumatic industrial rubber tires. The landing strut shall have a footprint of sufficient area to prevent the strut from sinking into soft ground.

3.11.2 Lifting and Tie-down Provisions: The pump assembly shall be provided with a lifting handle at the proper location and height to permit one man to easily wheel the unit over firm level ground. The pump assembly shall also be equipped with lifting and tie-down eyes suitable for hoisting and for securing the unit during air transport. The tie-down eyes shall be capable of holding the pump assembly when the acceleration forces specified in Military Specification MIL-A-8421 are applied.

3.12 Dimensions: The overall outside dimensions of the vacuum pump complete with wheels, and handles shall not exceed 45 inches in length, 36 inches in width and 36 inches in height.

3.13 Weight: The weight of the complete pump assembly shall not exceed 400 pounds.

3.14 Finishes and Protective Coatings

3.14.1 Exposed Parts and Surfaces: All exposed metal parts or surfaces shall be treated as specified herein.

3.14.1.1 Corrosion-Resistant Metal Parts and Surfaces: Metal parts and surfaces which are fabricated from corrosion-resistant materials need not be given protective coatings if such coatings will interfere with their proper functioning.

3.14.2 Chemical Treatment and Priming

3.14.2.1 Ferrous Materials: Ferrous parts or surfaces shall be cleaned and given a chemical treatment in accordance with Grade 1 of Specification MIL-C-490. A primer coat conforming to Specification TT-P-636 shall then be applied.

3.14.2.1.1 All exposed ferrous parts, such as screws, bolts, nuts, washers,

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etc. , shall be parkerized, bonderized, cadmium plated, or treated as specifically approved by the procuring activity. Cadmium plating shall be in accordance with Specification QQ-P-416, Type II or Type III, as applicable, and of a class that is adequate to achieve the degree of protection required.

3.14.2.2 Aluminum Materials: Aluminum parts and surfaces shall be cleaned in accordance with Specification MIL-S-5002. A corrosion-preventive chemical film conforming to Specification MIL-C-5541 followed by a primer coat conforming to Specification MIL-P-8585 shall *then* be applied.

3.14.2.3 Magnesium Materials: Magnesium base alloy parts and surfaces shall be cleaned and chemically treated in accordance with Specification MIL-M-3171. Two coats of primer conforming to Specification MIL-P-8585 shall then be applied.

3.14.2.4 Other Surfaces: Other types of surfaces shall be cleaned and chemically treated in accordance with good commercial practice. A coat of primer conforming to Specification MIL-P-8585 shall then be applied.

3.14.3 Finish Coats

3.14.3.1 Navy Equipment: Final treatment of exposed surfaces shall be the application of two coats of glossy enamel conforming to Specification TT-E-489. The color shall be gray conforming to color 16187 of Federal STD-595, unless otherwise specified. Markings shall be gloss black.

3.14.3.2 Marine Corps Equipment: Paint treatment shall be semi-gloss rust-inhibiting enamel conforming to Specification TT-E-485. Color shall be Marine Corps Green No. 23. Markings shall be black.

3.14.3.3 Alternate Finish: When approved by the procuring activity, use of commercial chemical (tricresyl phosphate) resistant finish on the pump, pump base motor, enclosure base, enclosure interior surfaces shall be acceptable. Colors shall be as specified in paragraphs 3.14.3.1 and 3.14.3.2.

3.15 Operating Instructions: Brief operating and precautionary instructions shall be suitably and permanently affixed on or near the control panel. The instructions shall be clear, concise, and adequate to enable operation without damage to the equipment or danger to the operator. The instructions shall include checking pump blank-off performance, connecting pump to LQOX tank, and checking the connection for leaks. The instruction panel shall be of such composition that exposure to oil, dirt, light, et cetera, will not fade it or cause it to become eradicated.

3.16 Identification of Product: Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.17 Workmanship

3.17.1 General: The vacuum pump, including all parts and accessories,

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shall be fabricated and finished in a workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring and riveting; alignment of parts and tightness of assembly screws and bolts; et cetera.

3.17.2 Dimensions: Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the vacuum pump or any of its components, they shall be held or limited accordingly.

3.17.3 Screw Assemblies: Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw, bolt, or threads.

3.17.4 Riveting: Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.17.5 Cleaning: The vacuum pump shall be thoroughly cleaned, and loose, spattered, or excess solder, metal chips, and other foreign material removed during and after final assembly.

4. QUALITY ASSURANCE PROVISION

4.1 Testing Responsibility: Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for official inspection and acceptance, The contractor shall utilize his own facilities or any commercial laboratory acceptable to the Government for conducting the specified tests. Inspection records of the examination and test shall be kept complete and available to the Government as specified in the contract order. All test equipment necessary for conducting tests in accordance with this specification shall be supplied by the Contractor.

4.1.1 Notification: The Contractor shall notify the Government Inspector and the Chief of the Bureau of Naval Weapons (SEQ,) 30 days prior to presenting the preproduction model for tests so that the Government Representative may witness all tests.

4.2 Classification of Tests: The inspection and testing of the pump unit (hereinafter referred to as the unit) shall be classified as follows:

(a) Preproduction Tests
(b) Acceptance Tests

4.2.1 Preproduction Tests: Preproduction tests shall consist of all the tests in Section 4.4 "TEST METHODS"

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4.2.2 Acceptance Tests: The acceptance tests shall consist of the following:

(a) Individual Tests
(b) Periodic sampling plans and tests

4.2.2.1 Individual Tests: Each vacuum pump shall be subjected to the following tests as described under 4.4 :

(a) Examination of product . . . . . .See 4.4.1
(b) Mechanical inspection . . . . . See 4.4.2
(c) Functional check . . . . . .See 4.4.3
(d) Operational check . . . . . See 4.4.4

4.2.3 Periodic Sampling Plans and Tests

4.2.3.1 Sampling Plan: One vacuum pump shall be selected at random from every fifty or fraction thereof produced and subjected to the tests specified in 4.4.7.

4.2.3.1.1 Rejection and Retest: When one vacuum pump selected from a production run fails to meet the specification, vacuum pumps still *on* hand or later produced shall not be accepted until the extent and cause of failure are determined.

4.2.3.1.2 Individual Tests May Continue: For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of vacuum pumps on hand or produced later shall not be made until it is determined that the vacuum pumps meet all the requirements of the specification,

4.2.3.1.3 Defects in Vacuum Pumps Already Accepted: The investigation of a test failure could indicate that defects may exist in vacuum pumps already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3 Test Conditions

4.3.1 Apparatus: Insofar as practicable, apparatus used in conjunction with testing prescribed herein shall be of laboratory precision type and shall be calibrated at intervals properly spaced to assure laboratory accuracy.

4.3.2 Instrumentation

4.3.2.1 Pressures: Wherever practicable, pressures and pressure differentials shall be measured by means of liquid manometers, Pressures exceeding the range of conventional liquid manometers shall be measured by means of properly calibrated Bourdon-tube gages. Such gages shall be of laboratory type having a dial diameter of not less than 4 inches,

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4.3.2.1.1 Pressures in the range from 1 to 100 mm Hg absolute shall be measured with closed-type manometers providing readings independent of prevailing barometric conditions.

4.3.2.1.2 Absolute pressures below 1 mm Hg shall be measured by the use of suitably calibrated instruments designed to read total pressures (including both permanent gases and condensable vapors). A certificate of instrument calibration to show compliance with 4.3.3 through 4.3.3.8 shall be included as part of the test report specified in paragraph 4.5.2.

4.3.2.1.3 Barometric pressures shall be measured by means of a properly calibrated mercurial or aneroid barometer.

4.3.2.2 Temperatures: Temperatures shall be measured by means of -appropriately located thermometers or thermocouples used with suitably calibrated potentionometers. Temperatures shall be expressed in degrees Fahrenheit.

4.3.2.3 Airflow

4.3.2.3.1 Airflow for free air displacement tests shall be measured with standardized nozzles or orifices as prescribed in ASME Code 19.5:4. The rate of airflow shall be reported in terms of the actual volume occupied under the existing ambient conditions.

4.3.2.3.2 When conducting tests to determine the pumping capacity under vacuum conditions, airflow into the evacuated container (if applicable) shall be measured by a method for which an error of less than 1 percent can be demonstrated. The rate of airflow under such conditions shall be reported in terms of the actual volume it would occupy under the pressure and temperature conditions existing in the evacuated container.

4.3.2.4 Revolutions per Minute: Whenever practicable, rotational speed shall be determined by means of a directly coupled positive counter which will actually count the revolutions for a period of not less than 1 minute. A combination unit which counts the revolutions while measuring the time required is preferred. Strobotachometers may be used provided calibration data can be furnished to attest to their accuracy. The use of-hand tachometers shall not be acceptable.

4.3.3 Tolerances

4.3.3.1 Data on absolute pressures measured in the range from 0 to 10 microns Hg shall be accurate to with 2 microns Hg.

4.3.3.2 Data on absolute pressures measured in the range from 10 microns Hg to 50 microns Hg shall be accurate to within 5 microns Hg.

4.3.3.3 Data on absolute pressure measured in the range from 50 microns Hg to 100 microns Hg shall be accurate to within 10 microns Hg.

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4.3.3.4 Data on absolute pressure measured in the range from 100 to 500 microns Hg shall be accurate to within 25 microns Hg.

4.3.3.5 Data on absolute pressures measured in the range from 500 microns to 1 mm Hg shall be accurate to within 50 microns Hg.

4.3.3.6 Data on absolute pressures measured in the range from 1 mm Hg to atmospheric pressure shall be accurate to within 0.1 mm Hg.

4.3.3.7 Data on airflow rates shall be accurate to within 1 percent.

4.3.3.8 Data on temperatures shall be accurate to within 0.5°F.

4.3.4 Vacuum Pump Lubricant: Commercial grade vacuum pump oil having a total vapor pressure of less than 1 micron Hg at +100°F shall be used as the lubricant and sealing fluid in all performance tests except as follows:

4.3.4.1 Type 2 tricresyl phosphate conforming to TT-T-656 shall be used as the vacuum pump lubricant and sealing fluid in the tests specified in 4.4.4.1, 4.4.6.4 and 4.4.7.

4.3.4.2 vacuum Pump Lubricant Consumption: Vacuum pump lubricant consumption shall be determined by measuring the quantity necessary to maintain a given level in the oil reservoir. Lubricant consumption shall be recorded on a volume basis.

4.3.5 Inspection, Servicing, and Adjusting: Inspections, servicing, and adjustments shall be performed as specified by the manufacturer unless such directions conflict with the requirements of this specification. Throughout the endurance tests, these services shall be performed during the shutdown periods.

4.3.6 Observations: During the progress of all tests, such characteristics as smoothness of operation, vibrations, ease of control, and the operation of any auxiliary equipment shall be carefully observed to determine whether the vacuum pump complies with the requirements specified herein.

4.3.7 Preparation. For Test: The vacuum pump shall be prepared for test by adjusting and servicing as specified by the manufacturer, except that the fuel and lubricating fluids specified herein shall be used.

4.3.8 Test Data

4.3.8.1 The following operational data shall be recorded at intervals not to exceed 15 minutes in performance checks and other short duration runs:

(a) Vacuum pump speed, rpm

(b) Absolute pressure of air or other gases entering the pump inlet, mm Hg or microns Hg, as applicable.

(c) Airflow entering vacuum pump, expressed in cfm or air or other gases measured under the inlet temperature and pressure conditions.

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4.3.8.2 The barometric pressure and temperature of the ambient air shall be recorded at intervals not to exceed 1 hour in performance checks and other short duration runs, and at intervals not to exceed 4 hours in endurance test runs,

4.3.8.3 The following operational data shall be recorded at intervals not to exceed 1 hour throughout the endurance test operational runs:

(a) Vacuum pump speed, rpm

(b) Absolute pressure of air or other gases entering the pump inlet, mm Hg or microns Hg, as applicable

(c) Temperature of air or other gases entering the pump inlet, degrees Fahrenheit.

4.3.8.4 Lubricant consumption data shall be reported at intervals not to exceed 24 hours in endurance test runs. Lubricant consumption shall be reported in terms of quarts consumed per hour of operation.

4.3.9 Operating Limits

4.3.9.1 Ambient Temperature: Insofar as practical, performance and endurante testing shall be conducted within ambient temperature limits of from 50° to 100°F.

4.3.9,2 Vacuum Pump Speed: The vacuum pump speed shall be maintained within ± percent of that recommended by the manufacturer.

4.3.10 Test Time: Intervals of endurance test time less than one complete 24-hour operational period terminated by a parts failure shall not be credited to the endurance test. Endurance test time shall be credited in 24-hour increments.

4.3.11 Running-In: The nature and extent of running-in shall be determined by the manufacturer. All necessary adjustments other than normal control adjustments shall be made during this run and shall remain undisturbed thereafter.

4.4 Test Methods

4.4.1 Examination of Product: The vacuum pump shall be inspected to determine compliance with the requirements specified herein with the respect to materials, workmanship, dimensions and marking.

4.4.2 Mechanical Inspection: A mechanical inspection shall be made of all components and parts of the vacuum pump, All pertinent data concerning conditions, defects of manufacture, damage in transit, and damage through use prior to test shall be recorded.

4.4.3 Functional Check: All mechanical parts of the vacuum pump shall be carefully checked for free and proper functioning.

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4.4.4 Operational Check: The vacuum pump shall be subjected to and shall satisfactorily complete the following tests before acceptance, and shall exhibit no malfunctions, leaks, excessive vibration, or other adverse operational characteristics.

4.4.4.1 Individual Proof Running: The vacuum pump shall undergo an operational proof run of not less than 12 hours utilizing tricresyl phosphate. The proof run time shall be divided as fellows and in the order listed:

(a) Four (4) hours of continuous operation at rated speed with the vacuum pump inlet and discharge connections open to atmospheric pressure.

(b) Four (4) hours of continuous operation at rated speed with the vacuum pump inlet pressure maintained between 50 and 500 microns Hg absolute.

(c) Four (4) hours of continuous operation at rated speed with the vacuum pump inlet hose completely blanked off except for a connection to the absolute pressure measuring device. The absolute pressure attained at the inlet shall be recorded at intervals not to exceed 15 minutes throughout this operation. Failure of the pump to obtain and maintain 10 microns Hg or less shall be cause for rejection.

4.4.5 Performance Mechanical Inspection and Tests

4.4.5.1 Initial Teardown Inspection: The vacuum pump used for testing shall be disassembled and inspected prior to further tests. All wearing parts and all parts subjected to high stresses in operation shall be accurately measured, ferrous parts shall be subjected to magnetic particle inspection in accordance with MIL-I-6868, and non-magnetic parts shall be inspected with flourescent penetrant (black light) in accordance with MIL-I-6866 for comparison with like inspections to be performed after completion of the tests. The vacuum pump shall be reassembled and subjected to the remainder of the tests.

4.4.6 Performance Demonstration: The vacuum pump shall be tested in accordance with the following procedures to determine its performance capabilities prior to endurance testing, Any malfunctions or unusual operating difficulties occuring during the tests shall be recorded. Failure to meet the performance requirements of Section 3,6 shall be cause for rejection.

4.4.6.1 Free Air Pumping Rate: The vacuum pump shall be operated at rated speed until its free air pumping rate, the number of actual cubic-feet measured at ambient conditions it will pump through per minute when both inlet and discharge are at atmospheric pressure, has been determined. The free air displacement shall be measured by means of a flowmeter connected to the inlet opening. The duration of this test shall be not less than 2 hours.

4.4.6.2 Ultimate Vacuum: The vacuum inlet hose flange shall be blanked off except for a connection to the absolute pressure measuring device and operated at rated speed with the discharge open to atmospheric pressure until the

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inlet pressure has remained stable for a period of not less than 1/2 hour. A check shall be made to insure that no flow of air or other gases is entering the pump inlet during this test. The vacuum pump shall have operated continuously for not less than one (1) hour prior to making the pressure check,

4.4.6.3 Pumping Rate at Low Pressures: The vacuum pump shall be operated at its rated speed. with the discharge open to atmospheric pressure until the rate in cfm (measured at inlet conditions of pressure and temperature) at which it is capable of evacuating a container maintained at 100 microns Hg absolute pressure has been determined, This test shall be conducted by one of the following methods:

4.4.6.3.1 Measured Leak Method: The vacuum pump shall be connected to a vacuum chamber of suitable size which has been tested and found to have no detectable leaks when operated at an absolute pressure of 100 ± 10 microns Hg. The vacuum pump shall be started and operated to evacuate the chamber to approximately 100 microns Hg absolute pressure. Air shall be permitted to leak into the chamber through a flow measuring device at a rate which will maintain the chamber pressure constant at 100 ± 10 microns Hg absolute pressure. The quantity of air permitted to leak into the chamber over a period of not less than 2 hours shall be determined, The quantity measured shall be converted to the actual volume at chamber pressure and temperature that was permitted to leak into the chamber per minute (actual pumping rate). The volumetric efficiency of the vacuum pump when operating at an absolute pressure of 100 microns Hg shall be calculated. The method of calculation for determining both the pumping rate and the volumetric efficiency shall be demonstrated,

4.4.6.3.2 Pressure Reduction Method: The vacuum pump shall be connected to a chamber of accurately known dimensions which has been tested and found to have no detectable leaks when operated at an absolute pressure of 100 ± 10 microns Hg. The vacuum pump shall be started and used to evacuate the chamber to a pressure less than 100 microns Hg absolute. The time required to evacuate the chamber from an absolute pressure of, for example, 200 to 50 microns Hg, shall be accurately determined by a series of not less than 10 individual tests. The internal volume of the vacuum chamber shall be not less than 10 cu.ft. From the known internal volume of the chamber plus its attachments and the length of time required to reduce the absolute pressure is indicated above, the pumping rate at a pressure of 100 microns Hg absolute shall be determined, The volumetric efficiency of the vacuum pump at an absolute pressure of 100 microns shall be calculated from this information. The methods of calculation for determining the pumping rate and the volumetric efficiency shall be demonstrated.

4.4.6.3.3 The connection from the vacuum chamber to the vacuum pump shall be 1-1/2 inches or larger iron pipe size.

4.4.6.4 Test With Tricresyl Phosphate: The tests specified in paragraphs 4.4.6.1, 4.4.6.2 and 4.4.6.3 shall be repeated using tricresyl

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phosphate as specified in 4.3.4.1 for the pump lubricant and sealing fluid. The tricresyl phosphate may be processed to remove water.

4.4.7 Endurance: The vacuum pump shall be subjected to an endurance test of not less than 504 hours. The lubricant and sealing fluid used throughout this test shall be tricresyl phosphate as specified in 4.3,4.1. The endurance test time shall be divided as follows:

(a) One hundred and twenty (120) hours of operation with the vacuum pump inlet and discharge connections open to atmospheric pressure.

(b) One hundred and twenty (120) hours of operation with the vacuum pump inlet completely blanked off except for a connection to the absolute pressure measuring device and the discharge open to atmospheric pressure. A check shall be made to insure that no flow of-air or other gases is entering the pump inlet during this test.

(c) Two hundred and sixty-four (264) hours of operation with the vacuum pump inlet pressure maintained between 50 and 500 microns Hg absolute.

4.4.7.1 Test Procedure: The endurance test time shall be accumulated by continuous operation at rated speed for periods of 12 hours under the specified conditions followed by shutdown periods of not less than 6 hours. Inspections, lubrication, minor adjustments, minor repairs, and preventive maintenance shall be accomplished during the shutdown periods.

4.4.8 Performance Recheck: Following the endurance test run and without intervening reconditioning other than normal preventive maintenance and servicing with lubricants, the vacuum pump shall be resubjected to the performance demonstration specified in paragraph 4.4.6. The performance at this time shall be not less than that specified in Section 3 for a new vacuum pump.

4.4.9 Final Teardown Inspection: The vacuum pump shall be disassembled as specified in 4.4.5.1 and a critical inspection made of all components to determine their operability and any damage or undue wear encountered during the testing.

4.4.9.1 The measurements taken under 4.4.5.1 shall be repeated for comparison with the data recorded prior to testing. Wear or distortion which exceeds the limits established by the manufacturer for new parts shall be cause for considering the affected parts as having failed to satisfactorily pass the tests.

4.4.9.2 Ferrous parts shall be resubjected to magnetic particle inspection (magnaflux) in accordance with MIL-I-6868 and shall exhibit no indicates of damage attributable to the tests, or they shall be considered as having failed the tests. Grind checks and subsurface indications of laps, seams, or other forging defects noted before the test and which have not increased in size shall not be cause for rejection.

4.4.9.3 Nonmagnetic casting, forgings, and other operating parts shall be

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subjected to inspection with fluorescent penetrant (black light) in accordance with MIL-1-6866, and shall exhibit no cracks or other defects, or shall be considered to have failed the tests.

4.4.10 Environmental: The unit shall be subjected to the following tests conducted in accordance with Specification MIL-E-4970. Failure of the pump to operate as required during or after each exposure, or failure of any items or assembly due to deterioration from exposure in this test, or formation of any undesirable condition which cannot be remedied by routine maintenance procedures, shall be cause for rejection.

4.4.10.1 Low Temperature - Procedure III: After the 48-hour exposure at -80°F, the ambient temperature in the cold room shall be increased to 30°F. After stabilization of temperatures at 32°F, starting procedures shall be accomplished. Failure of the motor to start and reach operating speed without the aid of external heat shall be cause for rejection. All gages, valves, meters, accessories, etc., shall be operated and shall perform satisfactorily in the 32°F ambient-temperature condition.

4.4.10.2 High Temperature - Procedure I: The purge unit shall be tested in accordance with procedure I of MIL-E-4970A.

4.4.10.3 Salt Fog: The salt fog test shall be conducted in accordance with MIL-E-4970 for a period of 50 hours. Operation of the unit is not required during this exposure.

4.4.10.4 Rain Tests - Procedure I: The unit shall be positioned on the test chamber floor in its normal operating position. After exposure to rain for one hour and while still in the rain test, the unit shall be started and operated at full rated output for a period of 30 minutes. Failure of the unit to start or operate satisfactorily during the rain test shall be cause for rejection. Any collection of moisture within the housing in excess of 1/8 inch deep or which could be detrimental to the operation of the tester after the rain test shall be cause for rejection.

4.5 Part Failures: If during the preproduction testing of a vacuum pump a part fails, that part shall be replaced with a redesigned part or one of a different material unless the procuring activity authorizes installation of a new part of the original design and material for one which failed due to faulty material or workmanship. The preproduction testing shall be considered completed when every major pact in the vacuum pump has passed the complete test. At the discretion of the procuring activity, redesigning and retesting may be required of any part which fails or indicates weakness after completing its preproduction test, but which is retained in the vacuum pump to complete testing of other parts.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packing

5.1.1 Level A: Each vacuum-pump shall be preserved in accordance with

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MIL-P-116, method II.

5.1.1.1 Prior to preservation, all oil shall be drained from the vacuum pump. All internal parts of the vacuum pump which may be subject to corrosion shall be thoroughly slushed with a compound conforming to MIL-C-6529.

5.1.2 Level C: Vacuum pumps shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing

5.2.1 Level A: Vacuum pumps preserved and packaged as specified in paragraphs 5.1.1 and 5.1.1.1 shall be packed in exterior shipping containers conforming to PPP-B-585, PPP-B-601, PPP-B-621, or MIL-B-10377. Insofar as practicable, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and shall contain identical quantities. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.2 Level B: Vacuum pumps preserved and packaged as specified in paragraph 5.1.1 shall be packed in domestic exterior containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, or MIL-B-10377. Exterior containers shall. be of minimum cube and tare consistent with the protection required. Insofar as practicable, exterior containers shall be of uniform shape and size and shall contain identical quantities. The gross weitht of each pack shall be limited to approximately 500 pounds. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C: Packages which require overpacking for acceptance by the carrier shall be packed in exterior shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet Consolidated Freight Classification rules or regulations of other common carriers, as applicable to the mode of transportation.

5*3 Physical Protection: Cushioning, blocking, bracing, and bolting as required shall be in accordance with JAN-P-100 except that waterproofing requirements for cushioning materials and containers shall be waived when the item is preserved, packaged, and packed for immediate use or when the drop tests of MIL-P-116 are applicable.

5.4 Marking: Interior and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

VACUUM PUMP, COMPOUND, ROTARY, MOTOR DRIVEN
12 CFM FOR LQOX TANK SERVICING

6. NOTES

6.1 Use: The vacuum pump covered by this specification is intended primarily for use in evacuating the insulating annular space of liquid oxygen and nitrogen storage tanks.

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6.2 Ordering Data: Procurement documents should specify the following:

(a) Title, Number, and Date of this specification.
(b) Colors required (see paragraph 3.14.3)
(c) Applicable level of preparation for delivery (See Section 5).

6.2.1 Preproduction Tests: One vacuum pump will be required as a preproduction sample to be subjected to the preproduction tests to determine compliance with the requirements of this specification.

6.3 Definition

6.3.1 Ambient Conditions: Ambient conditions shall be understood to mean the dry-bulb temperature and the barometric pressure of the air approaching the pump inlet connection.

6.4 Requirements For Descriptive Literature: Descriptive literature as specified below must be furnished as a part of the bid and must be received before the time set for opening bids. The literature furnished must be identified to show the item in the bid to which it pertains. The descriptive literature is required to establish for the purpose of bid evaluation and award, details of the products the bidder proposes to furnish as specified below:

(1) Dimensioned outline drawing of the vacuum pump assembly showing side view, rear view, and top view. This drawing shall also show all access doors, lifting and tie-down provisions, and electrical connections.

(2) A phantom side, end and top view of the vacuum pump showing the location of all major components, controls and piping.

(3) A drawing of the pump inlet piping showing the oil check device, the blank-off valve, the gaging provisions, and the filters.

(4) A bill of material, typewritten on bond paper giving the following information:

(a) Vacuum pump manufacturer, model number, manufacturer’s rating in c.f.m.

(b) The oil check device manufacturer's name, model number, and port size.

(c) The blank-off valve manufacturer’s name, model number, and port size.

(d) The thermocouple gage, manufacturer’s name, model number and scale range.

(e) The manufacturer’s name, model number, and rating for each

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of the following electrical devices:

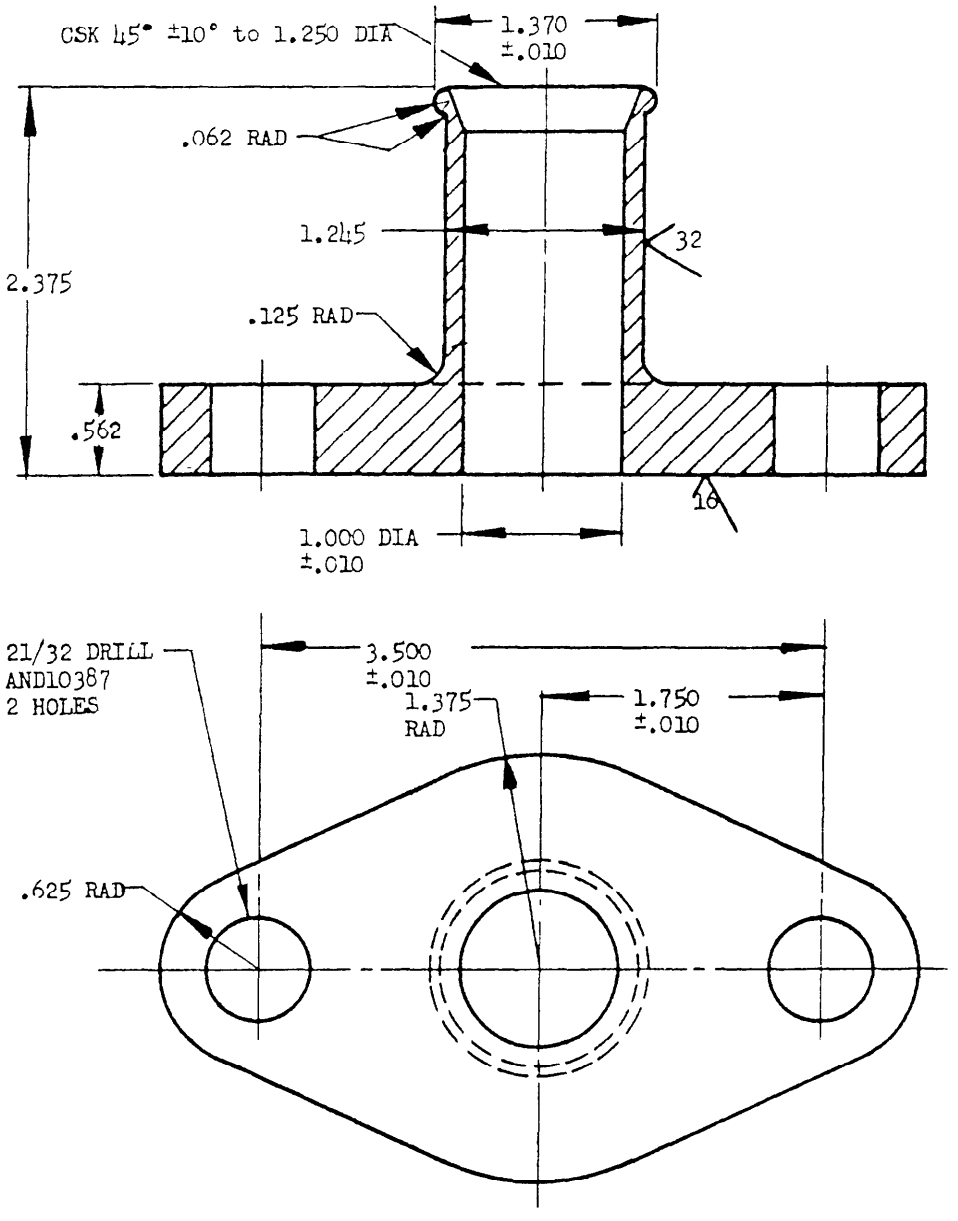
(1) Pump motor
(2) Pump motor starter
(3) Phase sequence control

(5) The manufacturer's catalog information specifically applicable to each component listed on the bill of material, (4) above, and which confirms the information given on the bill of material.

Failure of the descriptive literature to show that the product offered conforms to this specification and other requirements of the invitation for bids will require rejection of the bid. Failure to furnish the descriptive literature by the time specified in the invitation for bids will require rejection of the bid, except that if the material is transmitted-by mail and is received late, it may be considered under the provisions for considering late bids, as set forth elsewhere in the Invitation for Bids.

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

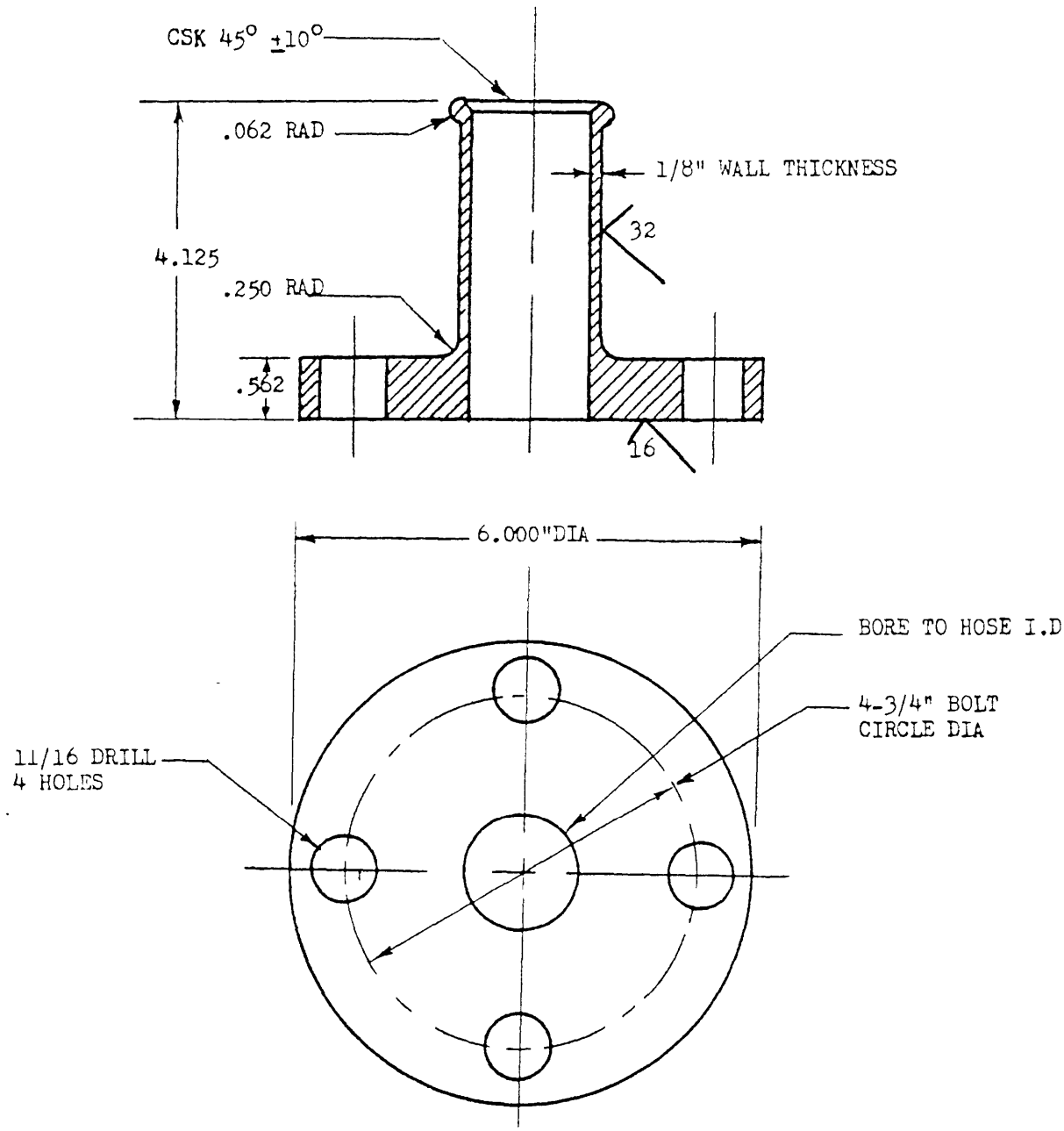
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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED,
TOLERANCES: DECIMALS ±.031.
SURFACE ROUGHNESS, MIL-STD-10.

FIGURE 1a. Flanged male fitting

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FIGURE 1b. SCALE - 1/2 SIZE

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